Question ID: b75f7812

Question

Maria plans to rent a boat. The boat rental costs $60 per hour, and she will also have to pay for a water safety course that costs $10. Maria wants to spend no more than $280 for the rental and the course. If the boat rental is available only for a whole number of hours, what is the maximum number of hours for which Maria can rent the boat?

Question ID: 208626df

Question

$$2\ell + 2w \leq 27$$

A rectangle has length ℓ and width w. The inequality gives the possible values of ℓ and w for which the perimeter of this rectangle is less than or equal to 27. Which statement is the best interpretation of $(\ell, w) = (8, 3)$ in this context?

Answer

A. If the rectangle has length 3 and width 8, its perimeter is less than or equal to 27.

B. If the rectangle has length 8 and width 3, its perimeter is less than or equal to 27.

C. If the rectangle has length 3 and width 8, its perimeter is greater than or equal to 27.

D. If the rectangle has length 8 and width 3, its perimeter is greater than or equal to 27.

Question ID: f01ef454

Question

A geologist estimates that the volume of a slab of granite is greater than 12.7 cubic feet but less than 15.7 cubic feet. The geologist also estimates that the slab of granite weighs 165 pounds per cubic foot of volume. Which inequality represents this situation, where x represents the estimated total weight, in pounds, of the slab of granite?

Answer

A. $165 - 15.7 < x < 165 - 12.7$

B. $165 + 12.7 < x < 165 + 15.7$

C. $165(12.7) < x < 165(15.7)$

D. $\frac{165}{15.7} < x < \frac{165}{12.7}$

Question ID: e744499e

Question

An elementary school teacher is ordering x workbooks and y sets of flash cards for a math class. The teacher must order at least 20 items, but the total cost of the order must not be over $80. If the workbooks cost $3 each and the flash cards cost $4 per set, which of the following systems of inequalities models this situation?

Answer

A. $x + y \geq 20$
$3x + 4y \leq 80$

B. $x + y \geq 20$
$3x + 4y \geq 80$

C. $3x + 4y \leq 20$
$x + y \geq 80$

D. $x + y \leq 20$
$3x + 4y \geq 80$

Question ID: 2869fe95

Question

For a 3-week period in a town in Illinois, the lowest recorded temperature was 31 degrees Fahrenheit $(°\mathrm{F})$ and the highest recorded temperature was $67°\mathrm{F}$. Which inequality is true for any recorded temperature t, in $°\mathrm{F}$, in this town for this 3-week period?

Answer

A. $t \geq 98$

B. $t \geq 67$

C. $31 \leq t \leq 67$

D. $t \leq 31$

Question ID: 84d0d07e

Question

A clothing store is having a sale on shirts and pants. During the sale, the cost of each shirt is $15 and the cost of each pair of pants is $25. Geoff can spend at most $120 at the store. If Geoff buys s shirts and p pairs of pants, which of the following must be true?

Answer

A. $15s + 25p \leq 120$

B. $15s + 25p \geq 120$

C. $25s + 15p \leq 120$

D. $25s + 15p \geq 120$

Question ID: 89541f9b

Question

Which of the following ordered pairs (x, y) satisfies the inequality $5x - 3y < 4$?

1. $(1, 1)$
2. $(2, 5)$
3. $(3, 2)$

Answer

A. I only

B. II only

C. I and II only

D. I and III only

Question ID: 915463e0

Question

Normal body temperature for an adult is between $97.8°F$ and $99°F$, inclusive. If Kevin, an adult male, has a body temperature that is considered to be normal, which of the following could be his body temperature?

Answer

A. $96.7°F$

B. $97.6°F$

C. $97.9°F$

D. $99.7°F$

Question ID: df32b09c

Question

Tom scored 85, 78, and 98 on his first three exams in history class. Solving which inequality gives the score, G, on Tom's fourth exam that will result in a mean score on all four exams of at least 90 ?

Answer

A. $90-(85+78+98)\leq 4G$

B. $4G+85+78+98\geq 360$

C. $\frac{(G+85+78+98)}{4}\geq 90$

D. $\frac{(85+78+98)}{4}\geq 90-4G$

Question ID: 563407e5

Question

A bakery sells trays of cookies. Each tray contains at least 50 cookies but no more than 60. Which of the following could be the total number of cookies on 4 trays of cookies?

Answer

A. 165

B. 205

C. 245

D. 285

Question ID: 2c121b25

Question

Valentina bought two containers of beads. In the first container 30% of the beads are red, and in the second container 70% of the beads are red. Together, the containers have at least 400 red beads. Which inequality shows this relationship, where x is the total number of beads in the first container and y is the total number of beads in the second container?

Answer

A. $0.3x + 0.7y \geq 400$

B. $0.7x + 0.3y \leq 400$

C. $\frac{x}{3} + \frac{y}{7} \leq 400$

D. $30x + 70y \geq 400$